

STORMWATER MANAGEMENT REPORT

OCTOBER 2022

LANDING ENCLAVE—EAST LOTS 1–4 AND OPEN SPACE LOT 5

*TAX MAP 37, PARCEL 66, GRID 06
F–22–063*

*PREPARED FOR:
LANDING ENCLAVE LLC
8318 FORREST ST., SUITE 300
ELLICOTT CITY MD 21043
410–997–0296*

***I HEREBY CERTIFY THAT THESE DOCUMENTS WERE PREPARED OR APPROVED BY ME
AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF
THE STATE OF MARYLAND, LICENSE NO. 26859, EXP DATE 08/08/2023***

SAMER ALOMER.

PROFESSIONAL ENGINEER

DATE



***MILDENBERG,
BOENDER AND ASSOC., INC.***

Engineers Planners Surveyors

*8318 FORREST STREET-SUITE 300
ELLICOTT CITY, MD 21043
(410) 997–0296 OFFICE
(410) 997–0298 FAX*

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INTRODUCTION:

The purpose of this report is to present the Stormwater Management design for 5740 Old Landing Road Property of Landing Enclave East, proposed Lots 1, 2, and 3, Parcel 66, Tax map 37, and demonstrate its compliance with the “Stormwater Management Act of 2007” as implemented in the 2000 Maryland Stormwater Management Design Manual Volumes I & II and its amendments and Howard County Design Manual Volume 1 including the 2017 revision.

This report includes a general description of the site, the proposed stormwater management which provides Environmental Site Design to the Maximum Extent Practicable, a discussion of why the proposed practice was chosen and the calculations that support the design of that practice and its compliance with the requirements, namely, that this Project will be subject to the 2007 revision of the Stormwater Management Act which is effective on May 4, 2010 and the County’s revised Design Manual Volume I.

GENERAL SITE INFORMATION:

The property is located on Old Landing Road approximately 300 feet north of Montgomery Road and 600 feet east of Landing Road. It consists of 2.36 acres. The property is zoned R-ED. The property may be located on the following maps:

1. TAX MAP: 37 - PARCEL : 66 – GRID 06
2. ADC MAP: 14, GRID H-6

The site drains from the southeast property corner to the northwest property corner. The site is composed of hydrologic soil Type "C".

The site contains wetlands, floodplains, and does not contain steep slopes. The site drains to an unnamed tributary which lies in the Rockburn branch watershed, as delineated by Howard County, which lies in the Patapsco River water shed (02-13-09-06).

Stormwater management is to be provided by three (3) proposed Micro-Bioretenction facilities. These Micro-Bioretenction facilities are designed to provide water quality treatment to the level of woods in good condition for "C" soils. The proposed micro-bioretenction facilities will be privately owned and maintained.

The proposed Micro-Bioretenctions will each discharge to a stable outfall. The Micro-Bioretenction areas and the site generally have been graded so that the proposed drainage divides mimic the natural divides that are part of the existing condition.

The ESD features have been designed in accordance with the latest Howard County and Maryland Department of the Environment (MDE) criteria to ensure to the maximum extent practicable that there are no deleterious effects on downstream areas.

Erosion and sediment controls have been designed in accordance with Howard Soil Conservation District (HSCD) criteria. The project has been designed so that no impacts to existing or proposed utilities or adjacent properties are anticipated. Soil excavation will be utilized as fill onsite wherever possible. Excess and / or unsuitable excavation is not anticipated but will be hauled off-site in an appropriate manner if encountered.

METHODOLOGY:

COMAR 23.17.02.02(2) requires that “All counties and their incorporated municipalities shall require that the planning techniques, non-structural and structural practices, and design methods specified in the Design Manual be used to implement ESD to the MEP...”

ESD is defined as “Environmental Site Design” (ESD) which means using small scale stormwater management practices, non-structural techniques, and better site planning to mimic natural hydrologic runoff characteristics and minimize the impact of land development on water resources.

MEP is defined as “Maximum Extent Practices” (MEP) which means designing a stormwater management system so that all reasonable opportunities for using ESD planning techniques and treatment practices are used and, only when absolutely necessary, a structural BMP are implemented.

Stormwater Management requirements for this project are being met entirely by Environmental Site Design (ESD) measures as described in Chapter 5 of the Design Manual.

The project has been designed so that no impact to existing or proposed utilities or adjacent properties is anticipated. The improvements and developed area will maintain the existing drainage pattern as closely as possible. No impact is proposed that would alter any natural flow patterns.

Soil excavation will be utilized as fill on site wherever possible. Excess and / or unsuitable excavation is not anticipated but will be hauled off-site in an appropriate manner if encountered.

Erosion and sediment control have been designed in accordance with Howard Soil Conservation District (HSCD) criteria and latest MDE standards and specifications. Additional changes to erosion and sediment control may be made based on their review of this Site Development Plan. It was determined that sediment traps were not necessary due to the fact that there is no concentrated flow and the slope to the super silt fence does not exceed 10 percent. Additional rows of super silt fence can be installed during construction, if required by the sediment control inspector.

For this project, non-structural ESD practices were used where possible. The required treatment depth for “Woods in Good Condition” for this site was determined in accordance with the 2007 revisions of Chapter 5 of the MDE 2000 Maryland Stormwater Design Manual Volumes I and II and the Howard County Stormwater Design Manual. The Micro-Bioretenion (M-6) was designed utilizing the formulas stated in Chapter 5 of the MDE manual. In order to provide safe conveyance for the runoff of the higher frequency storms, additional 6” high of storage is provided in the proposed Micro-Bioretenion facilities.

Per the manual, if the ESD target is met using the proposed non-structural BMP's, then the requirements for water quality volume (WQv), Recharge volume (Rev), and channel protection volume (Cpv) are met as well. Therefore, this development does not create an adverse impact on the adjacent properties.

In this analysis, the three (3) proposed houses shown are 2,000 square feet each.

The three (3) Micro-bioretentions (M-6) will be privately owned and maintained by the property owner.

The following are responses to the General Performance Standards set forth in the Maryland Department of the Environment (MDE) 2000 Stormwater Design Manual, Volume I, Chapter I:

- | | |
|-----------------|---|
| Standard No. 1: | The site has been designed to minimize the generation of stormwater and maximize pervious areas for stormwater treatment. |
| Standard No. 2: | Stormwater treatment has been provided for this development, however, it does not discharge directly to a jurisdictional wetland or water of the State. |
| Standard No. 3: | The Micro-Bioretentation has been designed to allow for as much recharge to the groundwater as possible. Runoff is designed to flow over grassed areas wherever possible. |
| Standard No. 4: | Water quality management is provided through the use of Micro-Bioretentation. |
| Standard No. 5: | By following the design standards in the MDE Stormwater Manual, the facilities are designed to provide removal of 80% of the total suspended solids load (TSS) and 40% of the average annual post development total phosphorous load. |
| Standard No. 6: | Since the site is not located in the Tiber watershed listed in Section 5.2.1.A of the Howard County Design Manual – Volume I, stormwater quantity management is not required above and beyond what is provided by the ESD measures designed for this project. |
| Standard No. 7: | Channel Protection storage volume (Cp) requirements have been met for this project through the design of the ESD measures using MDE and Howard County methodology. |
| Standard No. 8: | This project will not discharge any runoff to any critical areas with sensitive resources. |

- Standard No. 9: Operation and Maintenance agreements will be executed with all the proposed homeowners either during or after the Supplemental Plan stage of the project.
- Standard No. 10: The BMP's (ESD measures) have been designed per Howard County and MDE criteria concerning pre-treatment.
- Standard No. 11: Since the existing impervious coverage of the site is less than 40 percent, redevelopment stormwater management criteria does not apply to this project.
- Standard No. 12: The proposed project is not an industrial use.
- Standard No. 13: The proposed use (single-family residential) is not considered to be a "hot spot".
- Standard No. 14: Stormwater Management practices for this project will be reviewed and approved by Howard County, Maryland prior to beginning construction. NPDES construction general permit coverage will be applied for after the Site Development Plan (SDP) stage of the project, if required.

COMPUTATIONS:

SWM Requirements:

Site area: 113,675 s.f.

Environmental area: 31,134 s.f.

Developable area = 82,541 s.f.

Determine ESD (Environmental Site Design) implementation Goals

A. Determine pre-existing conditions (Woods in Good Condition)

Group 'A' Soils	:	0%	(Target CN = 38)
Group 'B' Soils	:	0%	(Target CN = 55)
Group 'C' Soils (113,731 sq. ft)	:	100%	(Target CN = 70)
Group 'D' Soils	:	0%	(Target CN = 77)

Determine composite RCN for "woods in good condition"

Pre-existing CN = 70

The target CN for "woods in good condition" is 70.

B. Determine Target PE using table 5.3:

Proposed impervious area:

Houses:	6,000 s.f.
Driveways:	6,280 s.f.
Sidewalk:	1,060 s.f.
Road widening:	1,390 s.f.

Total impervious = 14,730 s.f.

Percent Impervious (Area of proposed impervious/ Area of Disturbance) =

$14,730 (100) / 82,541 = 17.8\%$

Using Table 5.3 with I = 18% and soil group C

Rainfall Target **Pe = 1"**

Lot 1:

SWM requirements for the proposed houses and the proposed driveway will be provided in Micro-Bioretenction (M-6) facility MB #1.

Micro-bioretenction #1

Drainage area to Micro-bioretenction #1: 11,500 s.f.

Impervious area: House: 2,000 s.f.

Driveway: 1,800 s.f.

Total: 3,800 s.f.

$I: (3,800)(100)/11,500 = 33\%$

$R_v: (0.009)(33)+0.05 = 0.347$

Calculate Required ESD Volume:

$(11,500)(1.0)(0.346)/12 = 333 \text{ c.f.}$

Min. surface area : (2% of drainage area) : 230 s.f.

Surface area provided: 250 s.f.

Ponding area provided: 420 s.f.

Ponding Depth: 9"

Side Slope: 3:1

Volume in ponding required (75% REQ.): 250 c.f.

Volume in ponding provided: 251 c.f.

Volume in gravel required (25% REQ.): 83 c.f.

Depth of gravel below underdrain: 12"

Volume in gravel provided: 100 c.f.

Total Volume provided: $254+100=$ **351 c.f.**

651 c.f. > 332 c.f.

Lot 2:

SWM requirements for the proposed houses and the proposed driveway will be provided in Micro-Bioretentention (M-6) facility MB #2.

Micro-bioretentention #2

Drainage area to Micro-bioretentention #1: 6,820 s.f.

Impervious area: House: 2,000 s.f.

Driveway: 2,320 s.f.

Total: 4,320 s.f.

I: $(4.320)(100)/6.820 = 63.3\%$

Rv: $(0.009)(63.3)+0.05 = 0.62$

Calculate Required ESD Volume:

$(6,820)(1.0)(0.620)/12 = 352 \text{ c.f.}$

Min. surface area : (2% of drainage area) :136 s.f.

Surface area provided: 220 s.f.

Ponding area provided: 400 s.f.

Ponding Depth: 9"

Side Slope: 3:1

Volume in ponding required (75% REQ.): 264 c.f.

Volume in ponding provided: 310 c.f.

Volume in gravel required (25% REQ.): 88 c.f.

Depth of gravel below underdrain: 12"

Volume in gravel provided: 88 c.f.

Total Volume provided: $310+88=$ **398 c.f.**

398 c.f. > 352 c.f.

Lot 3:

SWM requirements for the proposed houses and the proposed driveway will be provided in Micro-Bioretenention (M-6) facility MB #3.

Micro-bioretenention #3

Drainage area to Micro-bioretenention #1: 7.370 s.f.

Impervious area: House: 2,000 s.f.

Driveway: 1,400 s.f.

Total: 3,400 s.f.

I: $(3,400)(100)/7,370 = 46.1\%$

Rv: $(0.009)(46.1)+0.05 = 0.465$

Calculate Required ESD Volume:

$(7,370)(1.0)(0.465)/12 = 286 \text{ c.f.}$

Min. surface area: (2% of drainage area) : 147 s.f.

Surface area provided: 180 s.f.

Ponding area provided: 390 s.f.

Ponding Depth: 12"

Side Slope: 3:1

Volume in ponding required (75% REQ.): 215 c.f.

Volume in ponding provided: 285 c.f.

Volume in gravel required (25% REQ.): 71 c.f.

Depth of gravel below underdrain: 12"

Volume in gravel provided: 72 c.f.

Total Volume provided: $285+72=$ **357 c.f.**

357 c.f. > 286 c.f.

Road Improvement:

SWM requirements for Old Landing Road widening, turnaround and sidewalk will be provided in 1 Micro-Bioretenction (M-6) facility MB #4.

Micro-bioretenction #4

Drainage area to Micro PLOT

Micro-retention #4: 45,200 s.f.

Impervious area: Driveway on Lot 4 750 s.f.
 Sidewalk: 1,060 s.f.
 Road widening: 1,390 s.f.
 Total: 3,200 s.f.

I: $(3,200)(100) / 45,200 = 7.1\%$

Rv: $(0.009)(7.1) + 0.05 = 0.114$

Calculate Required ESD Volume:

$(45,200)(1.0)(0.114) / 12 = 430 \text{ c.f.}$

Micro-bioretenction #4A and 4B

ESDv required for each MB: $430 / 2 = 215 \text{ c.f.}$

Min. surface area: (2% of drainage area) : 452 s.f.

Surface area provided: 452 s.f.

Ponding area provided: 660 s.f.

Ponding Depth: 9"

Side Slope: 3:1

Volume in ponding required (75% REQ.): 160 c.f.

Volume in ponding provided: 417 c.f.

Volume in gravel required (25% REQ.): 53 c.f.

Depth of gravel below underdrain: 6"

Volume in gravel provided: 90 c.f.

Total Volume provided: $417 + 90 = \underline{507 \text{ c.f.}}$

507 c.f. > 215 c.f.

Total Volume provided: $(507)(2) = 1,014 \text{ c.f.}$

SWM PRACTICES SUMMARY CHART:

AREA	PROPOSED PRACTICES	REQUIRED ESDv	PROVIDED ESDv	REQUIRED Pe	PROVIDED Pe
LOT 1	MICRO-BIORETENTION	332 C.F.	465 C.F.	1.0"	1.4"
LOT 2	MICRO-BIORETENTION	352 C.F.	442 C.F.	1.0"	1.26"
LOT 3	MICRO-BIORETENTION	286 C.F.	423 C.F.	1.0"	1.5"
ROAD IMPROVEMENT	MICRO-BIORETENTION	430 C.F.	1,014 C.F.	1.0"	2.36"
TOTAL:		1,400 C.F.	1,330 C.F.	1.0"	1.09"

CONCLUSIONS:

This report demonstrates that the proposed ESD practices (M-6) address the required treatment of the development of the proposed houses and driveways. The ESD methods chosen for this project are believed to be the best possible solutions to provide stormwater management.

The proposed Stormwater Management design meets all the requirements stated in the 2000 Maryland Stormwater Management Design Manual Volumes I and II as Amended in 2007 and Howard County design manual.

Per the manual, if the ESD target is met using the proposed non-structural BMP's, then the requirements for water quality volume (WQv), Recharge volume (Rev), and channel protection volume (Cpv) are met as well. Therefore, this development does not create an adverse impact on the adjacent properties.

By providing the Micro-Bioretenention (M-6) facilities #1, #2 and #3, the entire target runoff for this development has been treated.

Percent Volume Method:

$$\text{Rev} = (S)(R_v)(A)/12$$

$R_v=0.374$; A = Drainage area = 0.259 acres; S = Soil specific recharge factor

Composite Soil Recharge Factor			
Soil Group	Recharge Factor (S)	Area (acres)	(Area)(S)
A	0.38	0	0.000
B	0.26	0	0.000
C	0.13	0.259	0.034
D	0.07	0	0.000
Totals		0.259	0.034
Composite (S)			0.13

$$\text{Rev}_{\text{req'd}} = (0.13)(0.374)(0.259)/12 = 0.00104 \text{ ac. ft.} = \underline{\mathbf{46 \text{ cu. ft.}}}$$

$$\text{Rev}_{\text{provided}} = (316 \text{ sf})(1.167')(0.4 \text{ void space}) = \underline{\mathbf{147 \text{ cu. ft.}}}$$

Micro-bioretentention #2

Drainage area to Micro-bioretentention #1: 6,257 s.f.

Total impervious area: 4,623 s.f.

I: $(4623)(100)/6257 = 73.8\%$

Rv: $(0.009)(73.8)+0.05 = 0.714$

Calculate Required ESD Volume:

$(6257)(1.6)(0.714)/12 = 595 \text{ c.f.}$

Surface area required: (min.2% of drainage area)

$6257*0.02= 125 \text{ s.f.}$

Surface area provided: 313 s.f.

Ponding area provided: 580 s.f.

Ponding Depth: 12"

Side Slope: 3:1

Volume in ponding required (75% REQ.): 446 c.f.

Volume in ponding provided: 447 c.f.

Volume in gravel required (25% REQ.): 148 c.f.

Depth of gravel below underdrain: 15"

Volume in gravel provided: 157 c.f.

Total Volume provided: $447+157= \underline{604 \text{ c.f.}}$

604 c.f. > 595 c.f. required.

Recharge Volume Requirements (Rev)

Percent Volume Method:

$$\text{Rev} = (S)(R_v)(A)/12$$

$R_v=0.714$; A = Drainage area = 0.143 acres; S = Soil specific recharge factor

Composite Soil Recharge Factor			
Soil Group	Recharge Factor (S)	Area (acres)	(Area)(S)
A	0.38	0	0.000
B	0.26	0	0.000
C	0.13	0.143	0.019
D	0.07	0	0.000
Totals		0.143	0.019
Composite (S)			0.13

$$\text{Rev}_{\text{req'd}} = (0.13)(0.714)(0.143)/12 = 0.00110 \text{ ac. ft.} = \textbf{48 cu. ft.}$$

$$\text{Rev}_{\text{provided}} = (254 \text{ sf})(1.167')(0.4 \text{ void space}) = \textbf{157 cu. ft.}$$

SWM PRACTICES SUMMARY CHART:

AREA	PROPOSED PRACTICES	REQUIRED ESDv	PROVIDED ESDv	REQUIRED Pe	PROVIDED Pe	REQUIRED Rev	PROVIDED Rev
PROPOSED LOT 2	M-6, MICRO-BIORETENTION	595 C.F.	604 C.F.	1.6"	1.6"	48 C.F.	157 C.F.

Micro-bioretentention #3

Drainage area to Micro-bioretentention #1: 4160 s.f.

Total impervious area: 3330 s.f.

I: $(3330)(100)/4,160 = 80\%$

Rv: $(0.009)(80)+0.05 = 0.770$

Calculate Required ESD Volume:

$(4,160)(1.6)(0.770)/12 = 427 \text{ c.f.}$

Surface area required: (min.2% of drainage area)

$4,160 \times 0.02 = 83 \text{ s.f.}$

Surface area provided: 275 s.f.

Ponding area provided: 509 s.f.

Ponding Depth: 12"

Side Slope: 3:1

Volume in ponding required (75% REQ.): 320 c.f.

Volume in ponding provided: 392 c.f.

Volume in gravel required (25% REQ.): 107 c.f.

Depth of gravel below underdrain: 12"

Volume in gravel provided: 110 c.f.

Total Volume provided: $392+110 = \underline{430 \text{ c.f.}}$

430 c.f. > 427 c.f. required.

Recharge Volume Requirements (Rev)

Percent Volume Method:

$$\text{Rev} = (S)(R_v)(A)/12$$

$R_v=0.770$; A = Drainage area = 0.095 acres; S = Soil specific recharge factor

Composite Soil Recharge Factor			
Soil Group	Recharge Factor (S)	Area (acres)	(Area)(S)
A	0.38	0	0.000
B	0.26	0	0.000
C	0.13	0.095	0.012
D	0.07	0	0.000
Totals		0.095	0.012
Composite (S)			0.13

$$\text{Rev}_{\text{req'd}} = (0.13)(0.770)(0.095)/12 = 0.00080 \text{ ac. ft.} = \underline{\underline{\mathbf{35 \text{ cu. ft.}}}}$$

$$\text{Rev}_{\text{provided}} = (173 \text{ sf})(1.0')(0.4 \text{ void space}) = \underline{\underline{\mathbf{110 \text{ cu. ft.}}}}$$

SWM PRACTICES SUMMARY CHART:

AREA	PROPOSED PRACTICES	REQUIRED ESDv	PROVIDED ESDv	REQUIRED Pe	PROVIDED Pe	REQUIRED Rev	PROVIDED Rev
PROPOSED LOT 3	M-6, MICRO-BIORETENTION	427 C.F.	430 C.F.	1.6"	1.6"	35 C.F.	110 C.F.

